



MARKET MODELS

ASSIGNMENT 1

MODEL SPECIFICATION

This report provides a descriptive analysis of the market for lemonade, where the analysis is based on data of six brands and six distribution chains in the lemonade market. The data consists of the volume of sales per brand, per week and per chain and its promotional feature, taken over a total of 8736 observations. Moreover, data on prices, weekly temperature, seasonal effects and Covid is used. This report concludes on a specification of an appropriate predictive marketing model for one of the brands in the data set.



5 December 2022

DESCRIPTIVE ANALYSIS OF THE MARKET FOR LEMONADE



The lemonade market is a competitive world. Lemonade producers make an effort to let their brand stand out, where retailers compete with each other in obtaining the highest revenue. As the sweet beverage is an everyday product which is easily accessed by any customer, this market provides us with the perfect principles to analyse and model the lemonade market.

Description of variables

The data set consists of six brands in six well-known Dutch supermarket formulas, with 1248 observations for chain and 1456 observations for brands. There are 208 weekly data points available, running for 4 consecutive years (2017-2020), from week 1 (in 2017) to week 208 (in 2020). For each brand combination, data is provided on the volume of sales in units sold, the price per unit, price per liter, base prices per unit and base prices per liter. All prices are given in euro's and the base prices indicate the price level of the product when there is no promotional price drop. Furthermore, the data contains a dummy variable for the occasion where there is a special outside-store attention for the brand. This feature could for example be an advertisement in a newspaper. At the individual store level, this variable is assigned the value of one for the event of 'Feature-only' and a zero value otherwise. The weighted distribution of this variable indicates that the store-level dummy variables have been weighted with the turnovers of these stores. Hence, this variable varies between zero, meaning that not a single store has a feature-only, and a value of one when all stores have a feature-only.

Similarly for the variable 'Display-only', a weighted distribution for this variable applies. This variable indicates that there is special inside-store attention for a brand, for example a temporary shelf in the one of aisles. At the individual store level, this variable is given a dummy value where a one is assigned to an observation with display-only and a zero value elsewhere.

The variable "FeatDisp" represents the combined feature and display, again with a weighted distribution figure. The variable takes on a dummy value at the store level, so a one value for the combined use of feature and display, and zero elsewhere.

Finally, external data was added to account for the average weekly temperature, average new COVID cases per week (concerning 2020) and Holiday as a dummy variable. The COVID variable is set to zero for all weeks prior to the first COVID case in week 165. The external weather data is extracted from the KNMI, the holiday data is extracted from the governmental data base (data.overheid.nl) and the COVID data is provided by the John Hopkins Coronavirus Research Center.

Brands Chains

Private label, Euro Shopper, Karvan Cevitam, Raak, Slimpie, Teisseire
Albert Heijn, Jumbo, Plus, Coop, Deen, Hoogvliet, Online

Summaries of variables

The provided variables regarding price, volume sales and the availability of in-store and outside-store promotion have characteristics as described in table 1.1. It can be observed that both prices indicators per liter are higher than prices indicators per unit, as most units will contain less fluid than one liter.

Furthermore, while the feature-only and feature and display variables range from zero to one hundred, we find that the maximum share of stores that are using the display-only feature is 90 percent.

Variable	Min	Max	Mean	Median	Standard Deviation
Unit Sales	1	741177	27051	12960	44333.88
Price PU	0.7	3.19	2.004	1.76	0.6
Price PL	0.73	9.36	2.872	2.94	0.9
Base Price PU	0.73	3.180	2.066	1.78	0.63
Base Price PL	0.73	4.940	2.961	3.030	0.92
Feature	0	100	7.424	0	23.7
Display	0	90.000	1.135	0	5.1
Feature & Display	0	100.00	3.628	0	16.1

Table 1.1: summary statistics of lemonade data

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From the summary statistics in table 1.2, one can conclude that brands as Albert Heijn, Jumbo and Plus have the highest volume of sales, where the information also points out that these chains have the most volatile sales records.

Table 1.3 provides insights in which brands are most frequently sold. According to the data, Raak and KarvanCevitam are the most popular brands, followed by EuroShopper and the private label.

Chain	Min	Max	Mean	Median
Albert Heijn	1978	741177	64593	45802
Jumbo	7272	218844	59984	58218
Plus	1785	181962	20203	18312
Coop	613	91796	11375	11240
Deen	516	65426	6540	4246
Hoogvliet	1	76197	8137	7152
TotalOnlineSales	545	96750	11749	8848

Table 1.2: summary statistics sales per chain

Brand	Min	Max	Mean	Median
PrivateLabel	3525	266583	26659	16286
EuroShopper	2374	58593	22079	13012
KarvanCevitam	4562	741177	56316	21642
Raak	1203	337502	32317	17999
Slimpie	1162	70125	12232	6354
Teisseire	1	196774	7427	2966

Table 1.3: summary statistics sales per brand

Data cleaning and creating new variables

Before any analysis could be performed, several data entries required cleaning. The brands and chains were transformed to categorical values instead of the character values that they had. These variables were given factors, for which we would like to refer to the section of elucidating the model further in this report.

The weekly variable was transformed to a numerical value, ranging from one to 208 weeks. Moreover, an ordered variable was created to specify in which quarter of the year, either of the four quarters, every observation took place.

A variable on the way of advertisement was created, which indicates the sum of feature only, display only and feature & display.

Subsequently, three other variables were created NewFeatOnly, NewDispOnly and NewFeatDisp. If the variable advertisement is above 100 which is the limit of promotion then a function was created to deal with this problem. Three different functions were created for the 3 variables (NewFeatOnly, NewDispOnly and NewFeatDisp). This was an effective solution to solve the problem of the promotions that the sum of the promotion variables should not exceed 100%.

To include the effect that a promotion may have on the volume of sales in the surrounding weeks, two variables were created. The dummy variable 'NextWeek' represents the occurrence of a promotion happening in the next week, containing a one value when the event is happening and a zero value otherwise. The dummy variable 'PreviousWeek' contains a one value if a promotion took place in the last week, and a zero value otherwise.

Then external data was used to create a few more variables.

The variable 'Covid' was constructed by filtering the Covid cases in the Netherlands only. Only the data that suited our 208 weeks were captured and the weekly average was calculated. This averaging eliminated the peaks in this variable on a daily level, which might be caused by the irregular publishing of Covid data. The weekly average of the Covid data will be used for further analysis.

Another variable was created by taking the weekly average of the maximum temperature, measured by a weather station in De Bilt, which is centrally situated in the Netherlands.

Lastly, a variable was created to account for the presence of a holiday in the Netherlands. The variable takes on a value of one if a holiday was present in that week. A zero value was assigned to that week in the absence of any holiday.

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Outliers and irregularities

It was observed that there were 128 missings in the variables of unit sales, price per unit, price per liter, base price per unit, base price per liter. Furthermore, the brand Euroshoppers had many observations where the unit sales were equal to zero, which could not be classified as a transaction. After omitting these non-sales, the size of the data set decreased to 7568 observations, and the analysis on the lemonade data continued.

The data set contained outliers in multiple variables, such as the unit sales, price per liter and price per unit. To identify these outliers, the interquartile range of these variables was calculated. For the price per liter variable, there were two observations marked as outliers as they were positioned above the allowed maximum value according to this method. To deal with these outliers, the values were truncated to the nearest acknowledged value.

The outliers of unit sales retained their value as this will help to calculate the changes of unit sales caused by promotional actions.

The moving average was used to deal with the outliers of the price per unit variable. This implies that a new variable was created that displays the average price for the current week, the week prior and the week past it.

Sales per chain and sales per brand

The scatterplot as depicted in graph 1.1 clearly indicates that the most sales of lemonade brands are made by Albert Heijn, Jumbo and Plus. These chains also have the most fluctuating sales records, where the other four chains have rather steady sales records.

Graph 1.2 can be used to compare the number of sales made by each brand. KarvanCevitam can be seen as the market leader; it has the highest sales record with Raak close behind. The other four brands have significantly lower amount of sales over the given time period.



Graph 1.1: Unit sales per chain



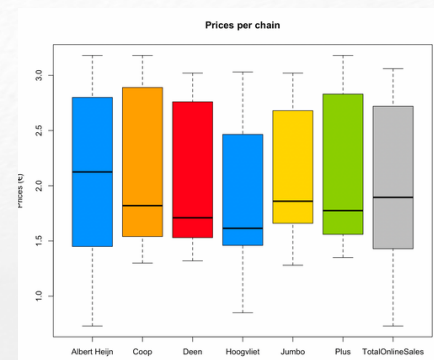
Graph 1.2: Unit sales per brand

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Prices per chain

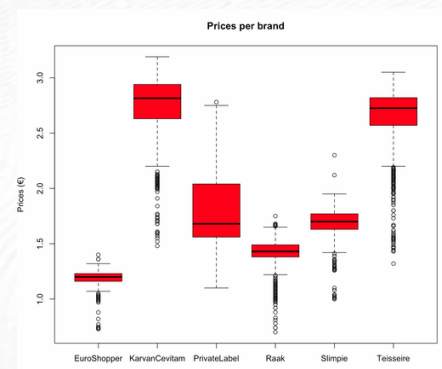
Graph 1.3 considers the range of prices per unit, split up by chains. Albert Heijn is said to ranks first as its median price lies well above the median of the others. The chain usually offers the most expensive products in the market of lemonade considering base prices. It is followed by Jumbo, Coop, and online sales chains. Finally, Deen and Hoogvliet can already be considered as low-priced chains.



Graph 1.3: Prices per chain

Prices per brand

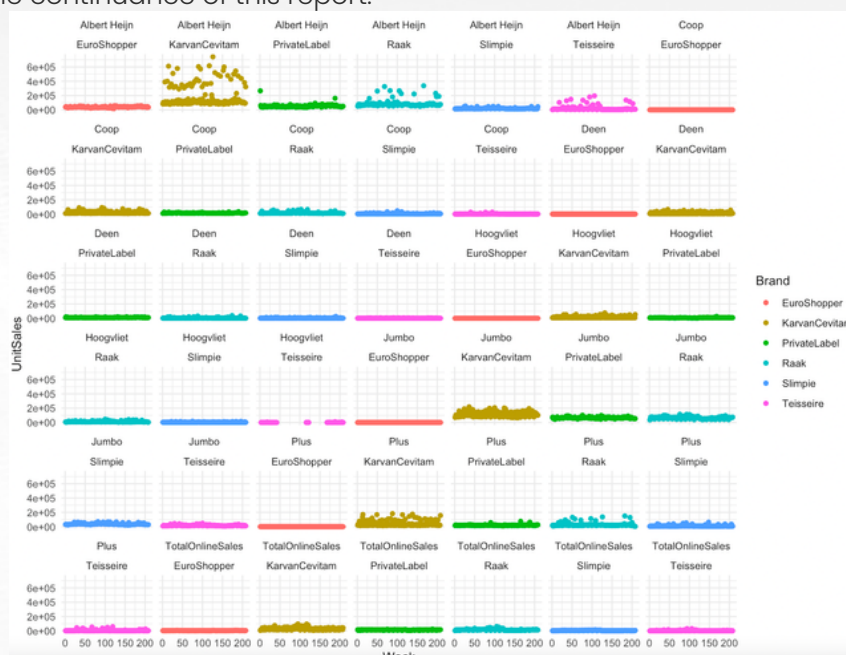
The visualisation about the prices per brand clearly depict that the prices of KarvanCevitam and Teisseire are the highest out of all brands, with a symmetrical distribution and with tall box plots, which indicates that the price differ per brand. EuroShopper and Raak are the cheapest brands and Slimpie and the PrivateLabel serve as the average group. Furthermore, outliers are identified in every brand in the box plot in graph 1.4.



Graph 1.4: Prices per brand

Aggregated visualisations

Graph 1.5 depicts the unit sales that occurred from the six different chains separately in combination with the six different brands, in the level of weeks. It is observed that brand Karvan Cevitam has the highest sales in the whole market chain, especially in the brand Karvan Cevitam where the difference between the other brands is significant. It is also worth pointing out that the remaining brands constitute a steady and similar unit of sales in the six different chains ranging over all weeks. From this, conclude that the brand KarvanCevitam is the most interesting to investigate in the continuance of this report.

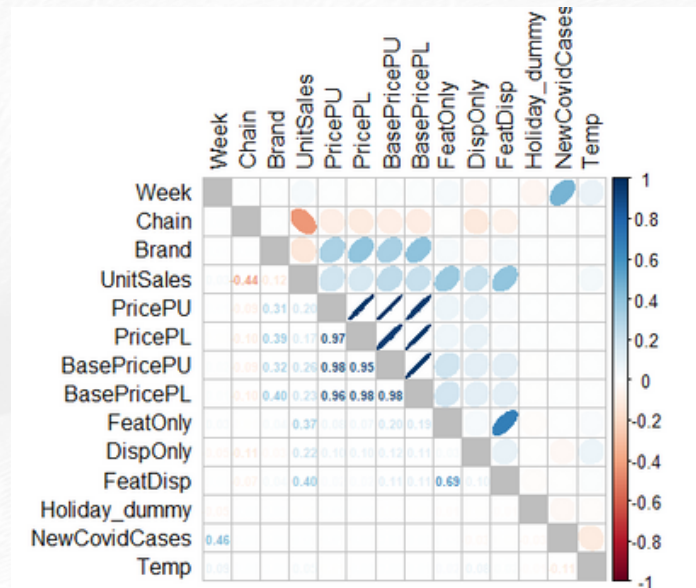


Graph 1.5: Volume of sales accross all brands and all chains

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Relational visualisations

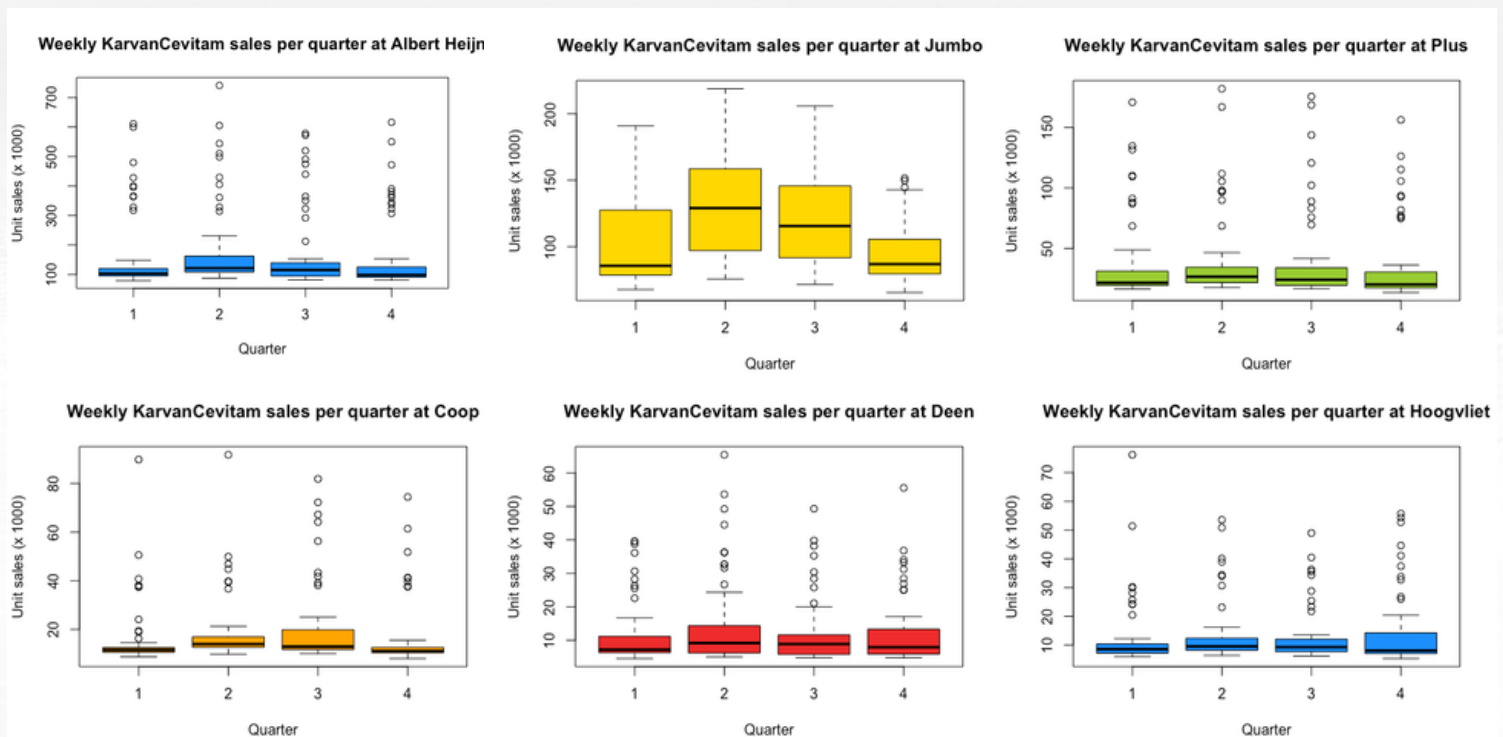
From graph 1.6 one can observe the positive correlations between all price variables. One can also observe the positive correlation coefficient of 0.69 of the feature-only variable and the variable on the combined feature and display. Lastly, a negative correlation coefficient of -0.44 is found on the relation of the volume of sales and the chains, which is due to the fact that the chains with the lowest index (Albert Heijn, Jumbo and Plus) yield the highest sales.



Graph 1.6: Correlation plot

Quarterly sales of KarvanCevitam

In the first part of graph 1.7, it can easily be seen that Albert Heijn has a few very big outliers regarding the sales in every quarter. This observation is depicted more clearly in graph 1.9. In the second plot, it is observed that the medians of sales in the quarters of Karvan Cevitam in Jumbo are quite different. This indicates that the median amount of sales is larger for the second and third quarters in comparison to the first and the fourth. The distribution is positive in the first and fourth quarter while in the remaining two it is symmetric. Furthermore, from the range of the sales it is concludes that Jumbo has the second largest amount of sales throughout the years and there are outliers identified in the fourth quarter.



Graph 1.7: Sales of KarvanCevitam per chain

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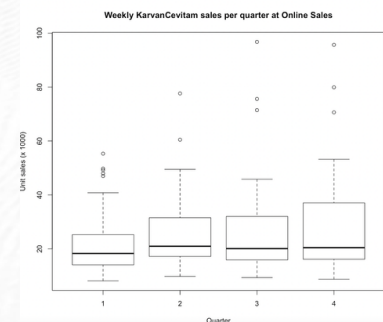
In the third plot, it is observed that all quarters have a small average and median amount of sales compared to the Albert Heijn and the Jumbo. The boxplots are comparatively short, which means that the unit sales are the same per quarter. Furthermore, outliers are identified in all of the quarters.

The fourth plot of graph 1.7 displays that the Coop has fewer sales than the Plus in every quarter and the quarters are positively distributed. The most sales are made in the second and third quarter and outliers are identified in all quarters.

In the fifth plot, it is observed that Deen has similar median sales in all quarters and the boxplots are positively distributed. The boxplots are tall which means that the sales are different per quarter. Furthermore, outliers are identified.

In the last plot really resembles to the sales of Deen, hence it can be concluded that Hoogvliet finds itself at a similar place in the market regarding volume of sales. It should not go unnoticed that Hoogvliet has the highest sales record in the fourth quarter and outliers are identified in all quarters.

In graph 1.8, it is observed that the boxplots in the second, third and fourth quarter have an almost identical median above 10000 unit sales, while the boxplot in the first quarter shows a lower median. The sales are positively distributed and the boxplots are comparatively tall, which indicates that unit sales differ per quarter. Furthermore, outliers are identified.



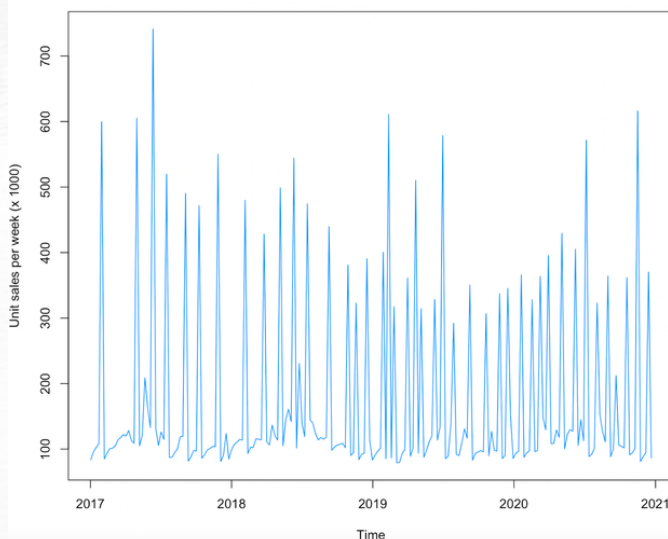
Graph 1.8: Online sales of KarvanCevitam

Sales of KarvanCevitam as time series

Graph 1.9 contains quite abstract peaks in sales that can be seen over time. The graph shows that there is a minimum of eight peak in a year and this clearly depicts the effect of weekly promotion on skyrocketing sales. Although seasonality effects could also be attributed to the graph, this information is not that obvious.

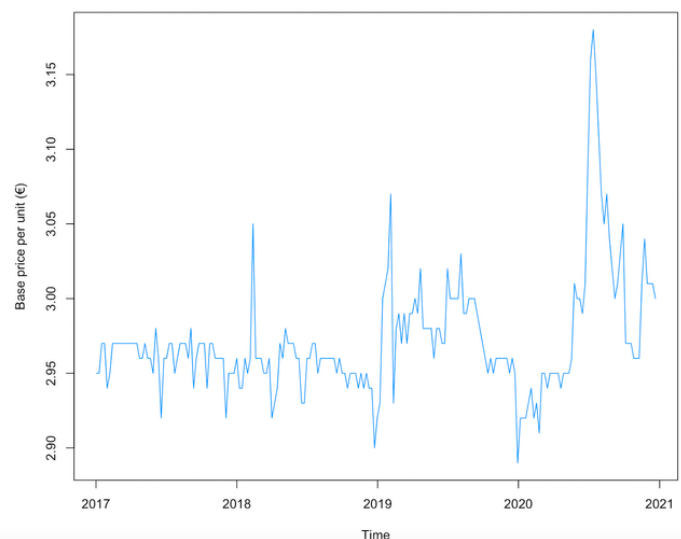
On graph 1.10, we see the range of price between 2,90 euros and 3,15 euros per unit. Specifically, we can clearly see that the base price per unit has been quite stable over the first two years, fluctuating around 2.95 euros per unit. Afterward two years, there is an upward trend of 0,10 euros per unit and more volatility. The peak identifies a price of 3,20 euros per unit.

Development of weekly unit sales of Karvan Cevitam over time



Graph 1.9: Time series plot of sales of KarvanCevitam

Development of base price of Karvan Cevitam over time



Graph 1.10: Time series plot of base price of KarvanCevitam

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Numerical analysis of relationships

To test the effect of seasonal influence on the sales of the lemonade data, an ANOVA test was conducted as depicted in graph 2.1. The results indicated that the p-value is less than 0.05 ($p < 0.00125$) and is statistically significant, thus the null hypothesis is rejected which indicates that the means of the quarters are not equal and the seasonal effect does influence the sales over time.

```
> anova_out<-aov(UnitSales~ Quarter, data= Lemonade)
> summary(anova_out)
              Df    Sum Sq   Mean Sq F value    Pr(>F)
Quarter         3  3.102e+10  1.034e+10   5.269 0.00125 **
Residuals      7564  1.484e+13  1.962e+09
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Graph 2.1: ANOVA test for seasonal influences on sales

In order to get a feel of our data, we run a simple linear regression for unit sales, using price per unit as our only independent variable, which can be found in graph 2.2. The result was slightly counterintuitive since it showed that the price coefficient is positive, meaning that an increase in price would increase the sales of lemonade. In order to investigate this further, we run the same linear regression for each chain.

The example that was chosen to make the interpretation for the regression was for the chain 'Albert Heijn' and the brand 'Karvan Cevitam', as depicted in graph 2.3. The result of the test indicates that the price per unit is significant because the p-value is less than 0.05 ($p < 2e-16$) but with a negative coefficient -460814. This indicates that the relationship is negative, meaning that when price increases with one unit, sales will decrease by 460814.

```
Call:
lm(formula = UnitSales ~ PricePU, data = Lemonade)

Residuals:
    Min       1Q   Median       3Q      Max
-30684 -21174 -14165   5155 713920

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  19745.6    1761.6   11.209  < 2e-16 ***
PricePU       3646.4     841.7    4.332  1.5e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 44280 on 7566 degrees of freedom
Multiple R-squared:  0.002474, Adjusted R-squared:  0.002342
F-statistic: 18.77 on 1 and 7566 DF, p-value: 1.496e-05
```

Graph 2.2: Simple regression on sales

```
Chain: Albert Heijn
Brand: KarvanCevitam

Call:
lm(formula = UnitSales ~ PricePU, data = x)

Residuals:
    Min       1Q   Median       3Q      Max
-216613 -17847  -2147   18370  222420

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  1468033    34328   42.77  <2e-16 ***
PricePU      -460814    12152  -37.92  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 49980 on 206 degrees of freedom
Multiple R-squared:  0.8747, Adjusted R-squared:  0.8741
F-statistic: 1438 on 1 and 206 DF, p-value: < 2.2e-16
```

Graph 2.3: Regression on the sales of Albert Heijn of KarvanCevitam

MODEL SPECIFICATION



Now that we have established a clear overview of the market for lemonade, distinguishing the volume of sales between brands and chains, all information is gathered to build a model to replicate this market. This model will use all variables of the previous section that are classified as influential in the market for lemonade.

The model of the lemonade market

We have decided to go with a multiplicative model. We believe that this type of model will be a better fit for our analysis because a multiplicative model offers certain advantages. The coefficients of the model are constant elasticities which allows for easier explanation of the model without needing to take into account the measurement scale of each variable. Also because the model is in logarithmic form, the range of each variable is reduced, the influence of outliers is reduced and more equal attention is given to each variable. On the other hand the logarithmic form makes the mathematics more complex and there is an inherent assumption in a multiplicative model that each variable interacts with each other. Despite those disadvantages, we believe that the pros outweigh the cons of using a model of that type.

Partial pooling or aggregation

The next decision we had to take into consideration was about pooling and aggregation. During the analysis, when trying to run a simple linear regression to see the relationship between sales and price across all chains, the model showed that sales and price have a positive correlation, meaning that when price increases, sales increase as well. Seeing how strange this is, we ran similar tests for each chain individually and saw that in all cases, sales and price have a negative correlation. Plus, the possible differences between customers of each chain pointed us towards a model that took the differences of each chain into consideration. Based on the above and the size of our dataset, we have decided not to do a unit by unit model, because we would have a small amount of data entries, and instead decided to follow a partial pooling model. This will enable us to use all of the dataset to determine the elasticities of each variable across all chains while simultaneously including different intercepts for each chain.

MODEL SPECIFICATION



Elucidating the model

Taking all of the above into consideration we came up with the following formula:

$$S_t = \alpha_i \cdot P_t^{\beta_1} \cdot B_t^{\beta_2} \cdot \beta_3^{F_t} \cdot \beta_4^{D_t} \cdot \beta_5^{FD_t} \cdot \beta_6^{G_{t-1}} \cdot \beta_7^{H_t} \cdot \beta_8^{V_{t-1}} \cdot T_t^{\beta_9} \cdot \beta_{10}^{H_t} \cdot \beta_{11}^{C_t} \cdot \gamma^{\alpha_2} \cdot \gamma^{\alpha_3} \cdot e^{\epsilon_t}$$

t = number of weeks

i = 1 for Albert Heijn, 2 for Coop, 3 for Deen, 4 for Hoogvliet, 5 for Jumbo, 6 for Plus, 7 for Total Online Sales

Variables

- S: Unit sales of Karvan Cevitam is our dependent variable.
- P: Price per Unit, measured in Euros, as price is an important factor of most customer decisions
- B: Average competitors price per week, measured in Euros
- NewFeatOnly2(F), NewDispOnly2(D), NewFeatDisp2(FD) as these variables state whether a promotion happened and that has a direct effect on consumer behavior
- Dummy variables to include the dynamic effects a promotion has on the week before and after it happens. Symbols in the model are: G for week before, V for week after
- Covid-19 cases per week. In 2020 the world was shocked with the emergence of the covid-19 pandemic which influenced us in many areas. We have included the average number of covid-19 cases per week as an indicator of the severity of the pandemic
- T: Temperature, measured in Fahrenheit to avoid negative numbers. As lemonade is a refreshing drink we assume that higher temperatures may influence the consumption of it, therefore by adding the weekly average temperature in our model we will be able to determine whether it has an effect or not
- H: Holidays are an occasion when people can come together which may alter the buying habits of people because they may need to purchase more or different products to host their friends or simply to support the gathering they are having. It will be interesting to see if those occasions have an effect on the sales of Karvan Cevitam
- Seasonality effects. Based on our analysis, there is a seasonality effect in the sales of Lemonade, therefore we will be including the seasonality effect in our model, a1=first quarter, a2=second quarter, a3=third quarter, a4=fourth quarter. The multiplier of seasonality will be labeled as γ .
- C: competitor promotion, when competitive brands have promotions that has an effect on the sales of Karvan Cevitam, therefore we want to examine how it affects it's sales
- Lastly, the error term in a multiplicative model is a multiplicative factor. This means that the value of the deterministic portion is multiplied by the error when the disturbance error is equal to zero. When it is positive, the multiplicative factor is greater than one, while when it is negative, the factor is smaller than one.

Literature review of the variables

Holiday: according to Horstmann (2020), 40% of all purchases are related to a specific occasion as it possesses a strong impact on shoppers' purchase behavior. The increase in sales is due to Christmas but also other holidays and seasonal feasts. Thus, holidays and seasonal feasts have a positive influence on sales.

Also M Zhang, X Huang, C Yang (2020) find a relationship between holiday effects and the volume of sales of consumption goods.

Temperature: according to Martínez-de-Albéniz and Belkaid (2021), temperature has a significant impact on sales. Notably cold drinks like lemonade, have higher average sales when the temperature is high.

Promotion: According to both P Chandon, B Wansink, G Laurent (2000) and KL Ailawadi, SA Neslin (1998) find evidence that promotion on everyday products increases its sales. Moreover, monetary promotions tend to lead to increased price sensitivity in the long run.

Multiplicative model: M Zhang, X Huang, C Yang (2020) use a multiplicative model to forecast the sales of consumer goods by measuring the holiday effects. This proves that a multiplicative model can be used to account for seasonal effects in forecasting sales of consumer goods.

In-store promotion (Display only): Vasanth Kiran (2012) argues in favor of open in-store media (ISM) which in a form of promotion have a result in the purchase decisions

Out-of-store marketing (Feature only):